

Explicit Formulas for Some Product Series in the White Noise Theory and Lévy Noise Analysis

H. M. Srivastava^{1,2,3,4,5,6,*} and Aleksandar Petojević⁷

¹Department of Mathematics and Statistics, University of Victoria,
Victoria, British Columbia V8W 3R4, Canada

²Department of Medical Research, China Medical University Hospital,
China Medical University, Taichung 40402, Taiwan

³Center for Converging Humanities, Kyung Hee University,
26 Kyungheedaero, Dongdaemun-gu, Seoul 02447, Republic of Korea

⁴Department of Applied Mathematics, Chung Yuan Christian University,
Chung-Li, Taoyuan City 320314, Taiwan

⁵Department of Mathematics and Informatics, Azerbaijan University,
71 Jeyhun Hajibeyli Street, AZ1007 Baku, Azerbaijan

⁶Section of Mathematics, International Telematic University Uninettuno,
39 Corso Vittorio Emanuele II, I-00186 Rome, Italy

E-Mail: harimsri@math.uvic.ca and harimsri@uvic.ca

ORCID ID: <https://orcid.org/0000-0002-9277-8092>

*Corresponding Author

⁷Faculty of Education, University of Novi Sad,
Podgorička 4, 25000 Sombor, Republic of Serbia

E-Mail: apetoje@uns.ac.rs

¹ORCID ID: <https://orcid.org/0000-0003-1491-165X>

Abstract

The main object of this article is to provide closed-form expressions for each of the following sums:

$$\mathcal{K}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N})^{-q\alpha} \quad \text{and} \quad \overline{\mathcal{K}}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N} + 1)^{-q\alpha},$$

which appear in the analysis of stochastic partial differential equations. Our main results express these sums as finite products of Gamma functions:

$$\mathcal{K}_q = \prod_{j=0}^{q-1} \Gamma\left(1 - \frac{\omega^j}{2}\right) \quad \text{and} \quad \overline{\mathcal{K}}_q = \frac{2^q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right),$$

where $\omega = e^{2\pi i/q}$ is a primitive q th root of unity and $q \geq 2$ is a natural number. The relevance and potential uses of explicit evaluation of

such product series in (for example) white noise theory and Lévy noise analysis is also indicated.

2020 Mathematics Subject Classification. Primary 60H15; Secondary 33B15, 11M06.

Key Words and Phrases. Stochastic partial differential equations; Kondratiev space; White noise analysis; Gamma function; Generalized stochastic processes; Kondratiev-Hida type spaces; Lévy noise; Fubini's theorem.

1. Introduction, Definitions and Preliminaries

We begin our present investigation by introducing the basic notation used in this work. Let $\mathbb{N}$ and $\mathbb{C}$ denote the sets of natural and complex numbers, respectively. Let $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. Also let $\mathcal{J}$ denote the set of all multi-indices, that is, sequences of the form:

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_i, \dots) \quad (\alpha_i \in \mathbb{N}_0),$$

with compact support (only finitely many components $\alpha_i \neq 0$).

As noted in [3, p. 35, Proposition 2.3.3], we have

$$\mathcal{K}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N})^{-q\alpha} = \sum_{\alpha \in \mathcal{J}} \prod_{k=1}^{\infty} (2k)^{-\alpha_k q} < \infty \quad (q > 1),$$

where, by definition,

$$(2\mathbb{N})^\alpha := \prod_{k=1}^{\infty} (2k)^{\alpha_k}.$$

This result is crucial for determining regularity conditions in spaces of generalized stochastic processes and is used to control norms in Kondratiev-Hida type spaces.

We define its counterpart as follows:

$$\overline{\mathcal{K}}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N} + 1)^{-q\alpha} = \sum_{\alpha \in \mathcal{J}} \prod_{k=1}^{\infty} (2k + 1)^{-\alpha_k q} \quad (q > 1),$$

where analogously we set

$$(2\mathbb{N} + 1)^\alpha := \prod_{k=1}^{\infty} (2k + 1)^{\alpha_k}.$$

This sum can be used in the analysis of stochastic partial differential equations driven by Lévy noise (see Chapter 5 in [3]), where similar infinite products involving odd integers appear.

Since each α has compact support, the product:

$$\prod_{k=1}^{\infty} (2k)^{-\alpha_k q}$$

is finite. We observe that the sum over all $\alpha \in \mathcal{J}$ can be decomposed into independent sums over each component α_k as follows:

$$\mathcal{K}_q = \sum_{\alpha_1=0}^{\infty} \sum_{\alpha_2=0}^{\infty} \cdots \prod_{k=1}^{\infty} (2k)^{-\alpha_k q}.$$

Since all terms are nonnegative, by Fubini's theorem we can interchange the order of summation and multiplication. Therefore, owing to the independence of the components, the following equality holds true:

$$\mathcal{K}_q = \prod_{k=1}^{\infty} \left(\sum_{\alpha_k=0}^{\infty} (2k)^{-\alpha_k q} \right).$$

The inner sum for each k is a geometric series with ratio $r_k = (2k)^{-q}$, for which $|r_k| < 1$ holds true for $q > 1$ and $k \geq 1$. Thus, clearly, we have

$$\sum_{\alpha_k=0}^{\infty} (2k)^{-\alpha_k q} = \sum_{n=0}^{\infty} r_k^n = \frac{1}{1 - r_k} = \frac{1}{1 - (2k)^{-q}},$$

which gives

$$\mathcal{K}_q = \prod_{k=1}^{\infty} \frac{1}{1 - (2k)^{-q}} \quad (q \in \mathbb{N} \setminus \{1\}). \quad (1)$$

The following identity is proved analogously:

$$\bar{\mathcal{K}}_q = \prod_{k=1}^{\infty} \frac{1}{1 - (2k+1)^{-q}} \quad (q \in \mathbb{N} \setminus \{1\}). \quad (2)$$

By combining (1) and (2), we obtain

$$\mathcal{K}_q \cdot \bar{\mathcal{K}}_q = \prod_{k=1}^{\infty} \frac{1}{1 - (2k)^{-q}} \cdot \prod_{k=1}^{\infty} \frac{1}{1 - (2k+1)^{-q}} = \prod_{n=2}^{\infty} \frac{1}{1 - n^{-q}}. \quad (3)$$

Finally, we recall the Gamma function $\Gamma(z)$ which is defined, for $z \in \mathbb{C} \setminus \mathbb{Z}_0^-$, by means of Euler's integral of the second kind:

$$\Gamma(z) = \begin{cases} \int_0^{\infty} e^{-t} t^{z-1} dt & (\Re(z) > 0) \\ \frac{\Gamma(z+n)}{\prod_{j=0}^{n-1} (z+j)} & (z \in \mathbb{C} \setminus \mathbb{Z}_0^-; n \in \mathbb{N}), \end{cases}$$

where, as usual, $\mathbb{C}$ denotes the set of complex numbers and

$$\mathbb{Z}_0^- := \mathbb{Z} \setminus \{0, -1, -2, -3, \dots\}$$

in the terms of the set $\mathbb{Z}$ of integers.

2. Main Results

In this section, we first state and prove the following theorem.

Theorem. *For every natural number $q \geq 2$, it is asserted that*

$$\mathcal{K}_q = \prod_{j=0}^{q-1} \Gamma\left(1 - \frac{\omega^j}{2}\right) \quad (4)$$

and

$$\overline{\mathcal{K}}_q = \frac{2^q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right), \quad (5)$$

where $\omega = e^{2\pi i/q}$ is a primitive q th root of unity.

Proof. We start from the following identity:

$$\prod_{k=1}^{\infty} \left(1 - \frac{z^q}{k^q}\right) = \prod_{j=0}^{q-1} \frac{1}{\Gamma(1 - \omega^j z)}, \quad (6)$$

valid for every $z \in \mathbb{C}$, where (as above) $\omega = e^{2\pi i/q}$ is a primitive q th root of unity.

The identity (6) is obtained by factorizing $z^q = \prod_{j=0}^{q-1} \omega^j z$ and applying the general form of the infinite product expressed via the Gamma function, as shown in [13, §12.13, p. 247]. Upon setting $z = \frac{1}{2}$, we obtain

$$\prod_{k=1}^{\infty} \left(1 - \frac{1}{(2k)^q}\right) = \prod_{j=0}^{q-1} \frac{1}{\Gamma\left(1 - \frac{\omega^j}{2}\right)},$$

and taking the reciprocal thereof, we obtain identity (4).

In order to present the proof of identity (5), we note that the left-hand side in (6) contains a term for $k = 1$ that equals 0 when $z = 1$, while the right-hand side includes $\Gamma(1 - z)$ which has a pole at $z = 1$. Therefore, to avoid the singularity, we consider the following product over all natural numbers $k \geq 2$:

$$\prod_{k=2}^{\infty} \left(1 - \frac{1}{k^q}\right) = \lim_{z \rightarrow 1} \frac{\prod_{k=1}^{\infty} \left(1 - \frac{z^q}{k^q}\right)}{1 - z^q} = \lim_{z \rightarrow 1} \frac{1}{1 - z^q} \prod_{j=0}^{q-1} \frac{1}{\Gamma(1 - \omega^j z)}.$$

As $z \rightarrow 1$, we use the known asymptotic of the Gamma function in the form:

$$\Gamma(1 - z) = \frac{1}{1 - z} - \gamma + O(1 - z),$$

where γ is Euler's constant. For $j = 0$ ($\omega^0 = 1$), we have

$$\frac{1}{\Gamma(1 - z)} = 1 - z + O((1 - z)^2).$$

For $j = 1, \dots, q-1$, since $\omega^j \neq 1$, it follows that $1 - \omega^j \neq 0$, so the function $z \mapsto \Gamma(1 - \omega^j z)$ is analytic in a neighborhood of $z = 1$. We also note that

$$1 - z^q = (1 - z)(1 + z + \dots + z^{q-1}) = q(1 - z) + O((1 - z)^2),$$

Therefore, we find that

$$\begin{aligned} \lim_{z \rightarrow 1} \frac{1}{1 - z^q} \cdot \frac{1}{\Gamma(1 - z)} &= \lim_{z \rightarrow 1} \frac{1}{q(1 - z) + O((1 - z)^2)} \\ &\quad \cdot \left(1 - z + O((1 - z)^2)\right) \\ &= \frac{1}{q}. \end{aligned} \tag{7}$$

For $j \neq 0$, $\Gamma(1 - \omega^j z)$ is regular in a neighborhood of $z = 1$. Thus, we get

$$\prod_{k=2}^{\infty} \left(1 - \frac{1}{k^q}\right) = \frac{1}{q} \prod_{j=1}^{q-1} \frac{1}{\Gamma(1 - \omega^j)}.$$

From here, by using equations (3) and (4), we obtain

$$\overline{\mathcal{K}}_q = \frac{q}{\sqrt{\pi}} \prod_{j=1}^{q-1} \frac{\Gamma(1 - \omega^j)}{\Gamma\left(1 - \frac{\omega^j}{2}\right)}. \tag{8}$$

Using the functional equation for the Gamma function, for each $j = 1, 2, 3, \dots, q-1$, we have

$$\Gamma(1 - \omega^j) = -\omega^j \Gamma(-\omega^j) \quad \text{and} \quad \Gamma\left(1 - \frac{\omega^j}{2}\right) = -\frac{\omega^j}{2} \Gamma\left(-\frac{\omega^j}{2}\right).$$

Therefore, we get

$$\frac{\Gamma(1 - \omega^j)}{\Gamma\left(1 - \frac{\omega^j}{2}\right)} = \frac{-\omega^j \Gamma(-\omega^j)}{-\frac{\omega^j}{2} \Gamma\left(-\frac{\omega^j}{2}\right)} = 2 \cdot \frac{\Gamma(-\omega^j)}{\Gamma\left(-\frac{\omega^j}{2}\right)}. \tag{9}$$

We now apply the duplication formula (that is, Gauss-Legendre theorem):

$$\begin{aligned} \Gamma(z) \Gamma\left(z + \frac{1}{2}\right) &= 2^{1-2z} \sqrt{\pi} \Gamma(2z) \quad \longrightarrow \quad \Gamma\left(-\frac{\omega^j}{2}\right) \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) \\ &= 2^{1+\omega^j} \sqrt{\pi} \Gamma(-\omega^j), \end{aligned}$$

which, upon solving for $\Gamma(-\omega^j)$, yields

$$\Gamma(-\omega^j) = \frac{\Gamma\left(-\frac{\omega^j}{2}\right) \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right)}{2^{1+\omega^j} \sqrt{\pi}}.$$

If we substitute this into the equation (9), we obtain

$$\begin{aligned} \frac{\Gamma(1 - \omega^j)}{\Gamma\left(1 - \frac{\omega^j}{2}\right)} &= 2 \cdot \frac{1}{\Gamma\left(-\frac{\omega^j}{2}\right)} \cdot \frac{\Gamma\left(-\frac{\omega^j}{2}\right) \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right)}{2^{1+\omega^j} \sqrt{\pi}} \\ &= \frac{2}{2^{1+\omega^j} \sqrt{\pi}} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) = \frac{1}{2^{\omega^j} \sqrt{\pi}} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right). \end{aligned}$$

We now return to the product in (8) given by

$$\begin{aligned} \prod_{j=1}^{q-1} \frac{\Gamma(1 - \omega^j)}{\Gamma\left(1 - \frac{\omega^j}{2}\right)} &= \prod_{j=1}^{q-1} \left(\frac{1}{2^{\omega^j} \sqrt{\pi}} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) \right) \\ &= \frac{1}{(\sqrt{\pi})^{q-1}} \cdot \frac{1}{\prod_{j=1}^{q-1} 2^{\omega^j}} \cdot \prod_{j=1}^{q-1} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right). \end{aligned} \quad (10)$$

Let us compute $\prod_{j=1}^{q-1} 2^{\omega^j}$. Thus, since ω is a primitive q th root of unity, we have

$$\sum_{j=0}^{q-1} \omega^j = 0 \quad \longrightarrow \quad \sum_{j=1}^{q-1} \omega^j = -1 \quad \longrightarrow \quad \prod_{j=1}^{q-1} 2^{\omega^j} = 2^{\sum_{j=1}^{q-1} \omega^j} = 2^{-1} = \frac{1}{2}.$$

Upon substituting this into the product:

$$\prod_{j=1}^{q-1} \frac{\Gamma(1 - \omega^j)}{\Gamma\left(1 - \frac{\omega^j}{2}\right)} = \frac{1}{(\sqrt{\pi})^{q-1}} \cdot 2 \cdot \prod_{j=1}^{q-1} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) = \frac{2}{\pi^{(q-1)/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right).$$

Returning next to (8), we have

$$\bar{\mathcal{K}}_q = \frac{q}{\sqrt{\pi}} \cdot \frac{2}{\pi^{(q-1)/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) = \frac{2q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right). \quad (11)$$

We now use the functional equation of the Gamma function to get

$$\Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right) = \left(\frac{1}{2} - \frac{\omega^j}{2}\right) \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) \quad \longrightarrow \quad \Gamma\left(\frac{1}{2} - \frac{\omega^j}{2}\right) = \frac{\Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right)}{\frac{1}{2} - \frac{\omega^j}{2}},$$

which, by substituting into (11), yields

$$\bar{\mathcal{K}}_q = \frac{2q}{\pi^{q/2}} \prod_{j=1}^{q-1} \frac{\Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right)}{\frac{1}{2} - \frac{\omega^j}{2}} = \frac{2q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right) \cdot \prod_{j=1}^{q-1} \frac{1}{\frac{1}{2} - \frac{\omega^j}{2}}.$$

In order to compute the product $\prod_{j=1}^{q-1} \left(\frac{1}{2} - \frac{\omega^j}{2} \right)$, we extracting the factor $\frac{1}{2}$ so that

$$\prod_{j=1}^{q-1} \left(\frac{1}{2} - \frac{\omega^j}{2} \right) = \frac{1}{2^{q-1}} \prod_{j=1}^{q-1} (1 - \omega^j).$$

It is known that

$$\begin{aligned} \prod_{j=1}^{q-1} (x - \omega^j) &= \frac{x^q - 1}{x - 1} \longrightarrow \prod_{j=1}^{q-1} (1 - \omega^j) = \lim_{x \rightarrow 1} \frac{x^q - 1}{x - 1} = q \\ &\longrightarrow \prod_{j=1}^{q-1} \frac{1}{\frac{1}{2} - \frac{\omega^j}{2}} = \frac{2^{q-1}}{q}. \end{aligned} \quad (12)$$

Finally, returning to the expression for $\overline{\mathcal{K}}_q$, we find that

$$\overline{\mathcal{K}}_q = \frac{2q}{\pi^{q/2}} \cdot \frac{2^{q-1}}{q} \prod_{j=1}^{q-1} \Gamma \left(\frac{3}{2} - \frac{\omega^j}{2} \right) = \frac{2^q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma \left(\frac{3}{2} - \frac{\omega^j}{2} \right).$$

This evidently completes the proof of the identity (5). □

Example. Based on the identities (4) and (5), if we set $q = 2, 3, 4$, then we have the following values for the functions $\mathcal{K}_q$ and $\overline{\mathcal{K}}_q$:

$$\begin{aligned} \mathcal{K}_2 &= \frac{\pi}{2} && \approx 1.57079632679489661922 \\ \mathcal{K}_3 &= \sqrt{\pi} \Gamma \left(1 - \frac{1}{2} e^{-2\pi i/3} \right) \Gamma \left(1 - \frac{1}{2} e^{2\pi i/3} \right) && \approx 1.17225040086287374339 \\ \mathcal{K}_4 &= \frac{\pi}{2} \Gamma \left(1 - \frac{i}{2} \right) \Gamma \left(1 + \frac{i}{2} \right) && \approx 1.07217758536212302042 \\ \overline{\mathcal{K}}_2 &= \frac{4}{\pi} && \approx 1.2732395447351626862 \\ \overline{\mathcal{K}}_3 &= \frac{8}{\pi^{3/2}} \Gamma \left(\frac{3}{2} - \frac{1}{2} e^{-2\pi i/3} \right) \Gamma \left(\frac{3}{2} - \frac{1}{2} e^{2\pi i/3} \right) && \approx 1.05394569866394706013 \\ \overline{\mathcal{K}}_4 &= \frac{16}{\pi^2} \Gamma \left(\frac{3}{2} - \frac{i}{2} \right) \Gamma \left(\frac{3}{2} + \frac{i}{2} \right) && \approx 1.01486566664329811677 \end{aligned}$$

Remark. From the product representations (1) and (2), the asymptotic behavior follows directly:

$$\lim_{q \rightarrow \infty} \mathcal{K}_q = 1 \quad \text{and} \quad \lim_{q \rightarrow \infty} \overline{\mathcal{K}}_q = 1.$$

This differs from some asymptotic estimates appearing in the literature, such as the following claim:

$$\lim_{p \rightarrow \infty} \mathcal{K}_p = \lim_{p \rightarrow \infty} C_p = 0,$$

which appeared recently in [2, pp. 6, 14, 15, 20, 22, 24].

3. Concluding Remarks and Observations

Motivated essentially by their occurrence in the analysis of stochastic partial differential equations, we have derived closed-form expressions for the sums:

$$\mathcal{K}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N})^{-q\alpha} \quad \text{and} \quad \bar{\mathcal{K}}_q := \sum_{\alpha \in \mathcal{J}} (2\mathbb{N} + 1)^{-q\alpha}$$

in the following form involving finite products of Gamma functions:

$$\mathcal{K}_q = \prod_{j=0}^{q-1} \Gamma\left(1 - \frac{\omega^j}{2}\right) \quad \text{and} \quad \bar{\mathcal{K}}_q = \frac{2^q}{\pi^{q/2}} \prod_{j=1}^{q-1} \Gamma\left(\frac{3}{2} - \frac{\omega^j}{2}\right),$$

where $\omega = e^{2\pi i/q}$ is a primitive q th root of unity and $q \geq 2$ is a natural number.

The relevance and potential uses of explicit closed-form evaluation of each of the above product series in (for example) white noise theory and Lévy noise analysis could possibly be reiterated by the fact that there is remarkably on-going interests in the white noise as well as Lévy noise researches from many different angles and many different viewpoints. For the ready reference of the interested readers, we have chosen to include citations of some of the recent developments of these subjects (see, for example, [4], [5], [6], [9], [10], [11] and [14] for the white noise theory, and [1], [7], [8], [12] and [15] for the Lévy noise analysis).

Funding: This research received no external funding.

Data Availability Statement: Data are contained within the article. Data sharing not applicable to this article as no data sets were generated or analyzed during the current study.

Conflicts of Interest: The authors declare that they have no conflicts of interest.

References

- [1] F. Bai, Z. Chen and X. Sun, Existence and approximation of measure attractors and invariant measures for McKean-Vlasov stochastic lattice system with Lévy noise, *J. Differential Equations* **453** (2026), Article ID 113784, 1–79.
- [2] S. Gordić, T. Levajković and Lj. Oparnica, Stochastic very weak solutions to parabolic equations with singular coefficients, *J. Math. Anal. Appl.*, **555** (2026), Article ID 130023, 1–30.
- [3] H. Holden, B. Øksendal, J. Ubøe and T. Zhang, *Stochastic Partial Differential Equations: A Modeling, White Noise Functional Approach*, Second Edition, Springer Nature Switzerland AG, Cham, Switzerland, 2010.

- [4] Y.-S. Hsu and C. Labbé, Construction and spectrum of the Anderson Hamiltonian with white noise potential on $\mathbb{R}^2$ and $\mathbb{R}^3$, *Probab. Math. Phys.* **7** (2026), 1–35.
- [5] Y. Jiang, Y. Wang and J. Xue, Parameter estimation of Burgers equations driven by white-colored noise, *Statist. Probab. Lett.* **227** (2026), Article ID 110553, 1–8.
- [6] R. Li and Y. Li, Transportation cost-information inequality for non-linear time-fractional stochastic heat equation driven by space-time white noise, *Statist. Probab. Lett.* **227** (2026), Article ID 110519, 1–8.
- [7] K. Mathiyalagan, N. S. Lakshmi, H. Zhang and N. Zhao, Boundary control of stochastic partial differential systems driven by Lévy noise, *IMA J. Math. Control Inform.* **42** (2025), Article ID dnafo41, 1–21.
- [8] S. Moualkia and F. S. S. Al-Nom, Approximate controllability of fractional differential equations driven by Markovian switching and Lévy noise with infinite delay, *J. Control Decis.* **13** (2026), 180–201.
- [9] P. H. Quan, N. Q. Huy, L. M. Triet and L. H. Phong, The backward problem for the nonlinear space-fractional diffusion equation with Gaussian white noise, *Kyungpook Math. J.* **65** (2025), 639–667.
- [10] K.-Y. Ren, Optical solitons, dynamic behaviors, and chaotic characteristics of the stochastic fourth-order nonlinear Schrödinger equation with white noise, *Wave Motion* **142** (2026), Article ID 103694, 1–15.
- [11] N. H. Tuan and T. N. Thach, Rayleigh-Stokes equations with space-time white noise: Existence, Hölder regularity, and parameter-continuity, *Potential Anal.* **64** (1) (2026), Article ID 34.
- [12] C. Wang, H. Shu, S. Huisheng, Y. Shi and X. Zhang, Parameter estimation for partially observed stochastic processes driven by Ornstein-Uhlenbeck processes with small Lévy noises, *Internat. J. Systems Sci.* **57** (2026), 167–175.
- [13] E. T. Whittaker and G. N. Watson, *A Course of Modern Analysis: An Introduction to the General Theory of Infinite Processes and of Analytic Functions; with an Account of the Principal Transcendental Functions*, Fourth Edition, Cambridge University Press, Cambridge, London and New York, 1927.
- [14] W. Zhao and X. Liu, Invariant measure of non-autonomous stochastic reaction-diffusion equations with infinite delay and additive white noise, *J. Math. Anal. Appl.* **558** (2026), Article ID 130453, 1–31.
- [15] J. Zhu and J. Zhai, Stochastic logarithmic Schrödinger equations driven by Lévy noise, *Commun. Pure Appl. Anal.* **26** (2026), 103–149.